Interpretable NN Bot Detection via Behavioral Features in the LLM Era

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INTRODUCTION

- Twitter hosts milions of accounts, and social spambots are becoming increasingly common among those accounts
- They are automated accounts that mimic real users and spread misinformation
- We built an ML model that flags bot accounts based on behavioral & profile features
- We also prioritized performing a feature analysis to understand the model's decision-making process

LITERATURE REVIEW

Cresci et al. 2017

- Concluded that those detection techniques are incapable of accurately detecting social spambots.

Yang et al. 2019

- Built a more capable detector, found that carefully selecting a subset of training data helps with the model's accuracy.

Chen et al. 2022

- Presented the largest dataset to date, evaluated 35 Twitter bot detection baselines against it.

RESEARCH OBJECTIVES

- Develop a feedforward neural network able to successfully identify bot accounts.
- Derive features based on columns in dataset that will improve the model's bot detection.

METHODOLOGY

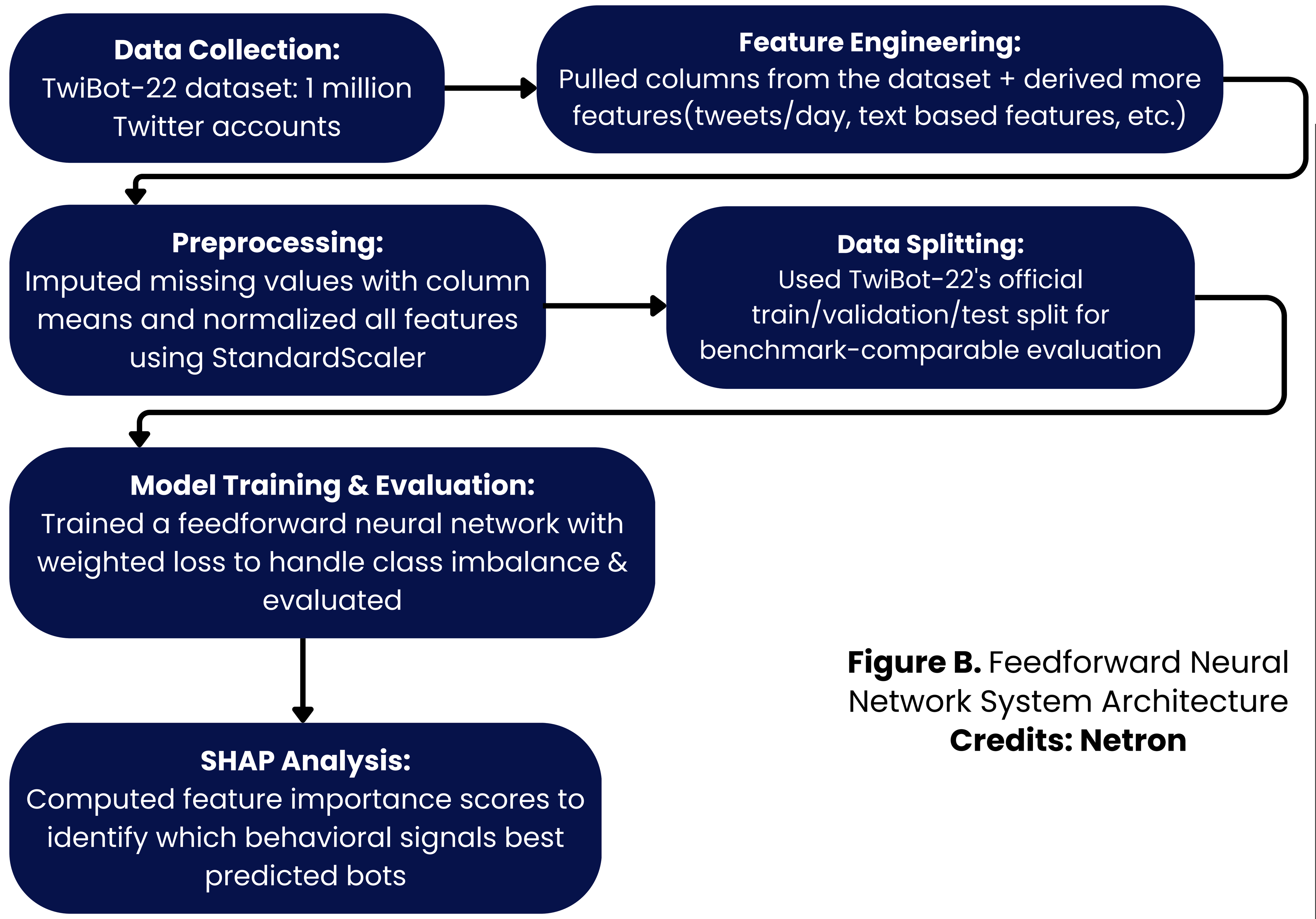

Figure A. Bot Detection Implementation Flowchart

Figure B. Feedforward Neural Network System Architecture
Credits: Netron

RESULTS

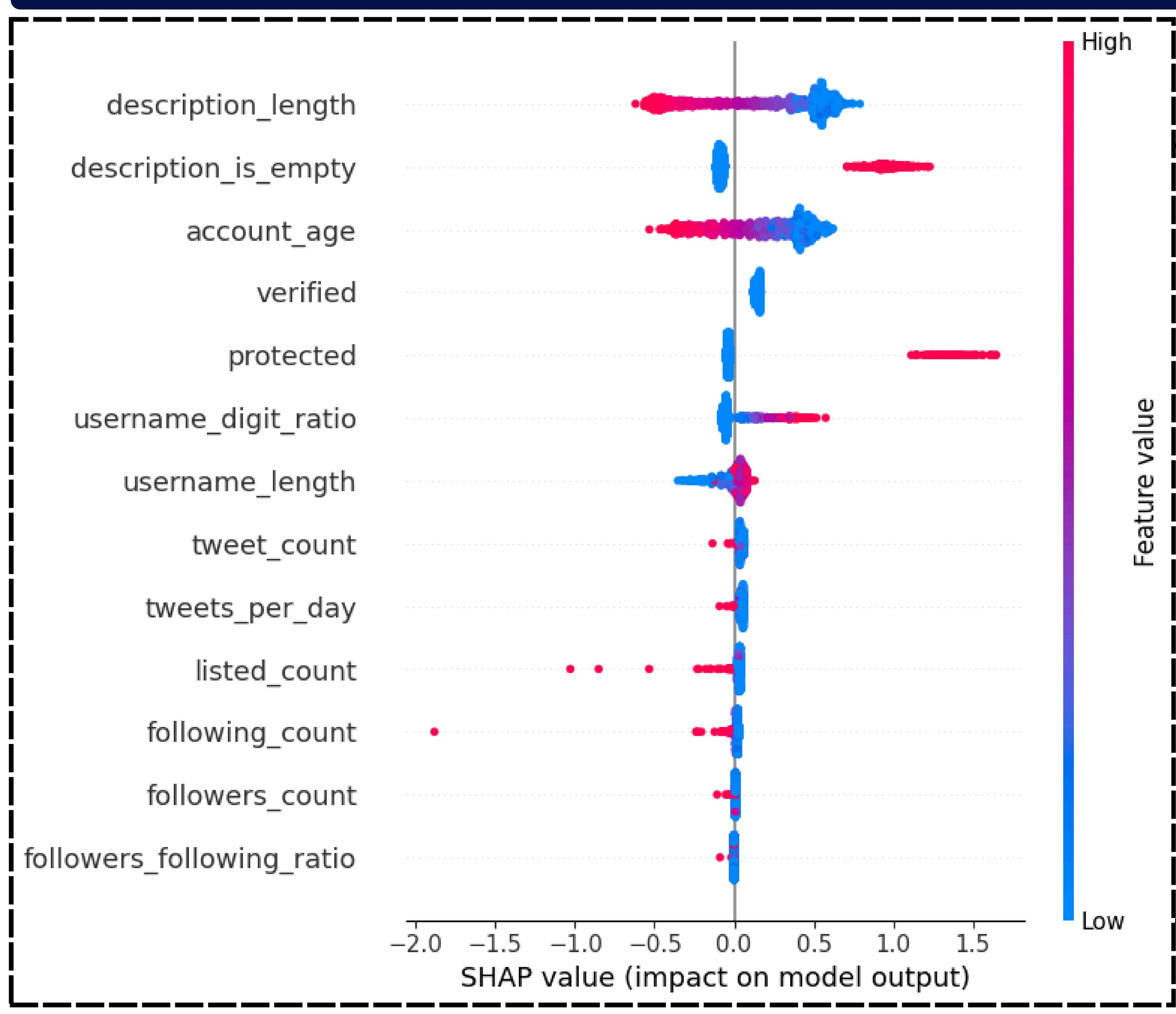

Figure C. SHAP Analysis Graph

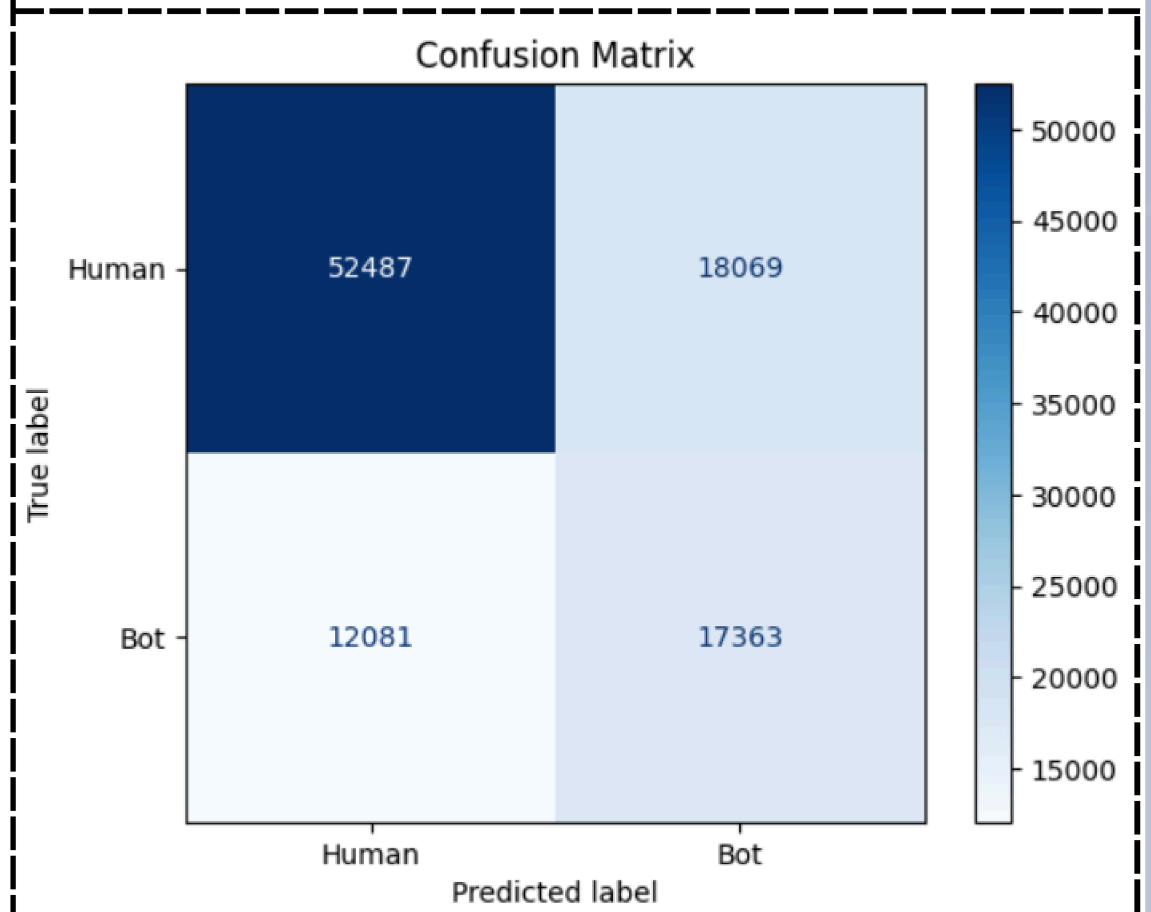

Figure D. Confusion Matrix

Figure E. Metrics Table

Accuracy:	Recall:	F1 Score:	ROC-AUC:
69.84%	59.84%	53.64%	71.74%

DISCUSSION

Test accuracy: 69.84%
ROC-AUC score: 71.73%- This shows that the model separated bots from humans accurately instead of randomly guessing
F1 score: 53.64%
Confusion matrix results: 52,487 humans detected and 17,363 bots detected

- Our model still makes mistakes when the bot accounts have behavior or profile features that are similar to real users.
- These results show that account-level features contain useful information.

CONCLUSION

Future Work: Continued development of new features and advancement of capabilities present to improve the model's detection capabilities. The creation of new data as LLM's advance to ensure the model continues being exposed to new types of social spambots.

Overall, the project showed that the account features can help identify which account is a social spambot with the output in either a 0(human user) or 1 (bot).

ACKNOWLEDGMENTS & REFERENCES

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